

9708/32/o/n/17

1-D:

- A.** The statement only talks about the allocation of resources, not about individuals' judgment of their well-being.
- B.** This option is also not directly related to the statement. The statement doesn't make any assumptions about individuals' behavior or rationality. It only describes a situation where resources are allocated in a particular way.
- C.** This option is not necessarily true based on the statement. While the resources are allocated efficiently, it doesn't mean that the distribution of income is socially optimal. The statement only talks about the allocation of resources, not about the distribution of income.
- D.** This is the correct answer. The statement defines Pareto efficiency, which means that resources are allocated in such a way that it's impossible to make one person better off without making someone else worse off. This is a definition of efficient allocation of resources because this implies that we are operating on the PPC.

2-A:

- ➔ Since, labor is a “variable factor”, a reduction in the number of staff by 6000, represents a decrease in the total variable cost
- ➔ By selling less efficient mines, fixed costs can be decreased. **Mines incur various fixed costs, such as administrative expenses, maintenance costs, and infrastructure upkeep. By selling less efficient mines, the company can eliminate or reduce these fixed overhead costs associated with those particular mines. This can include costs related to management salaries, office space, utilities, and other general operating expenses.**
- ➔ Since, average cost is “total cost/output”, and because of a fall in total variable costs and total fixed costs, it can be said that total cost falls, this means that there is a “**fall**” in average cost.

3-C:

- A – When external costs exceed external benefits, this represents net external cost.
- B – Private costs cannot exceed social costs.
- C _ When social costs exceed private costs, this represents the existence of external costs. This is why key C is correct.
- D – When social costs exceed social benefits, this represents net social costs.

4-D:

- D - When water is free (method 1), consumers tend to use it excessively because they don't bear the cost of each additional unit. However, when they have to pay per liter (method 2), they become more mindful of their consumption because they now bear the cost of each additional unit.

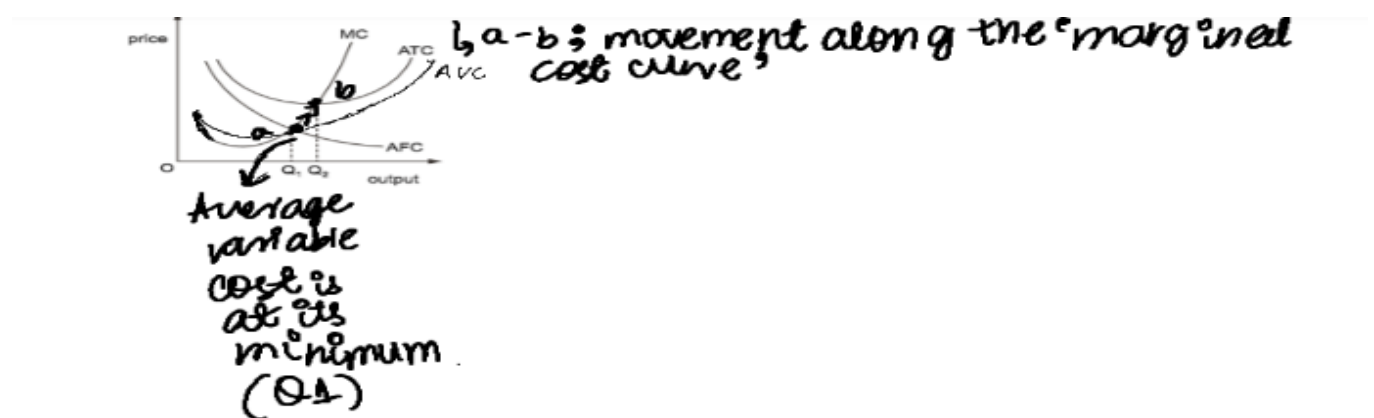
The observation that water consumption decreases when charged per liter implies that consumers are responding to the price signal. This is because the additional cost of each liter of water now outweighs the additional benefit (the marginal utility) they derive from consuming it. As a result, they reduce their consumption. This explains **that water is subject to diminishing marginal utility**.

A -The observation suggests the opposite: water is an economic good which is scarce therefore prices have been charged.

B- Water is not a free good; instead it is an economic good which is scarce in nature.

C - While water is a necessity, the observation suggests that pricing it per liter helps manage consumption, which is a reasonable restriction.

5-B:



A – As the firm increases its output from Q_1 to Q_2 , **the average variable cost increases**, as it was at “ Q_1 ”, whereby the **average variable cost** was at its lowest when the marginal cost curve intersected the average cost curve. **This can be seen in the diagram above.**

B – An increase in output from “ Q_1 to Q_2 ” will result in a movement along the marginal cost curve from “a to b”, which represents a rise in marginal cost. This occurs as a result of a fall in the “marginal product of labor”, as a result of which, the “law of diminishing returns” sets in. As more units of labor are employed, there comes a point where the efficiency of labor starts to diminish. The **marginal product of labor begins to decline**, indicating that each additional unit of labor contributes less to the total output, and as a result of which, **total product** increases at a **diminishing rate or decreasing rate**.

The diminishing marginal product of labor is a manifestation of the **Law of Variable Proportions**. It highlights that in the short run, increasing the quantity of labor beyond a certain point will eventually lead to **diminishing returns**.

This occurs due to factors such as **limited space, inadequate supervision, or bottlenecks in the production process**, which constrain the efficiency and effectiveness of additional labor.

This is why, key B, becomes correct.

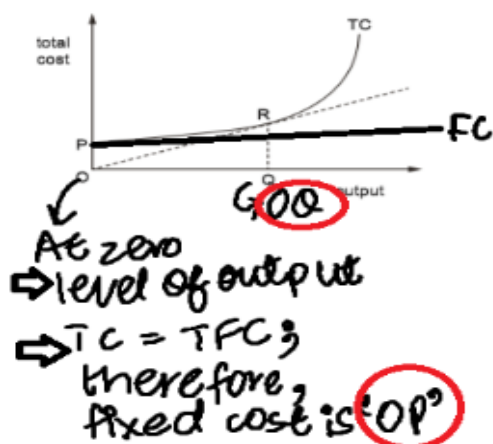
C- We are not asked about a **movement** along the **Average total cost curve**; instead, we are interested in a movement along the marginal cost curve.

D – We do not have sufficient information to understand the impact on profits.

6-D:

Horizontal integration occurs when firms in the same industry merge at the same stage of production. This explains why key D is correct.

7-C:



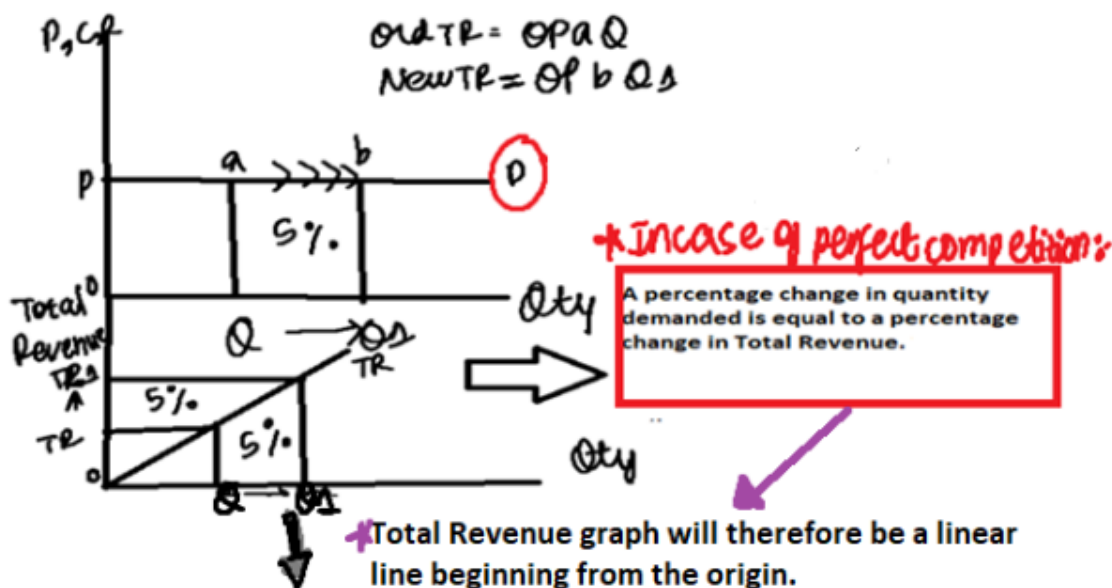
Average fixed cost $\Rightarrow$

$$= \frac{\text{Total fixed cost}}{\text{Quantity}}$$

$$= \frac{OP}{00} \rightarrow \text{option C}$$

8-C:

C -



C – The diagram explains why key C will be correct.

A – Here quantity demanded remains unchanged but since total revenue can increase, this represents that there must be a positive relationship between price and total revenue. This is only the case of perfectly inelastic demand.

B – Here total revenue remains unchanged which is only seen in case of unitary elastic demand where $PED=1$

D -

D - Option **D** represents the case of "imperfect competition", whereby there is a downward-sloping demand curve. This can be seen in the diagram below.



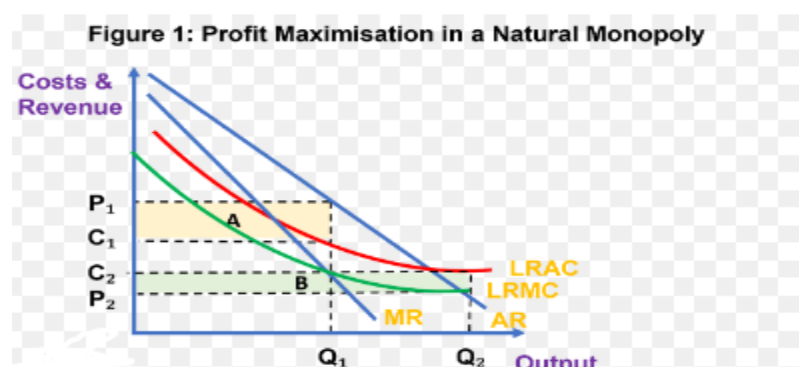
9-A:

A: A natural monopoly occurs when a single firm can supply the entire market demand at a lower cost than multiple firms. This is often the case with public utilities like water, electricity, and gas, which have high fixed costs (e.g., building a power plant or water treatment facility) and so the average fixed costs keep falling throughout the range of output which means that there are decreasing average costs over the entire range of output demanded. This means that as the firm produces more, its average cost per unit decreases, making it more efficient than multiple smaller firms.

B. While legal barriers can contribute to a monopoly, they aren't the defining characteristic of a natural monopoly.

C. Control over raw materials can be a barrier to entry which can result in a monopoly, but it's not the primary reason for a natural monopoly.

D. A patent, that is a legal barrier, can give a firm a temporary monopoly status, but it's not the same as a natural monopoly.



10-B:

B: When a firm in an imperfectly competitive industry increases its advertising, it can lead to an increase in demand for the firm's product. However, if rival firms respond by increasing their own advertising efforts, this can lead to a fall in the original firm's profits as consumers may switch to the rival firms. This is because the increased advertising by rivals can offset the gains from the original firm's advertising.

A. Diseconomies of scale would affect the firm's costs, but profits can still rise after advertising even if the industry is experiencing dis-economies of scale.

C. Price inelasticity would not the firm's profits in response to advertising. In order to assess the impact of advertising efforts on revenue, we need information about promotional elasticity of demand.

D. If the firm is gaining customers at the expense of rivals, its profits should increase, not fall.

11-C:

In a situation where a dominant firm acts as a price leader, the small firms are essentially price-takers, meaning they have no control over the market price. To maximize profits, the small firm will operate at the profit maximizing level of output where $MC=MR$, but **since the price remains unchanged as the small firms have no control over the level of market**, this represents that $AR(\text{price})=MR$. This can be seen through the table below:

price	qty	TR	AR	MR
5	1	5	5	5
5	2	10	5	5
5	3	15	5	5
5	4	20	5	5
5	5	25	5	5

$$\begin{aligned} MC &= MR \\ MC &= AR \rightarrow MR = AR \\ &\quad (\text{price}) \end{aligned}$$

This is why option C will be correct.

12-D:

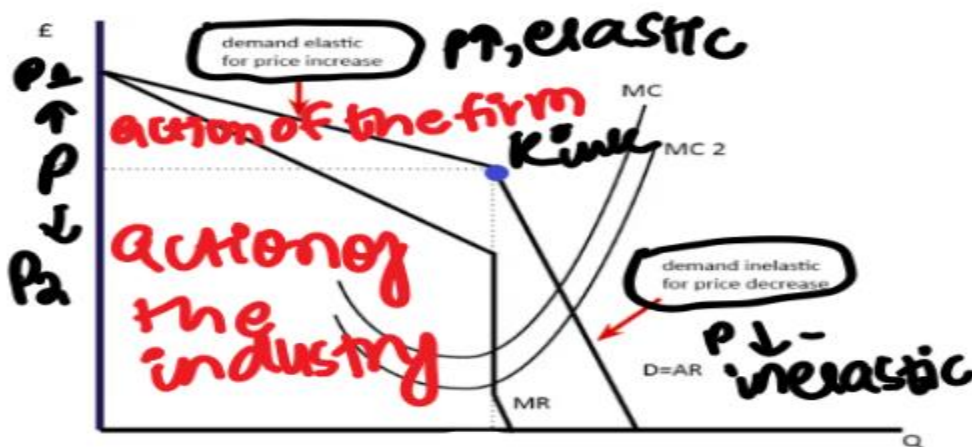
D - This situation describes a classic prisoner's dilemma scenario, where two firms (Firm X and its rival) have to make a decision without knowing what the other will do. The possible outcomes and payoffs are being given. The prisoner's dilemma arises because each firm's dominant strategy is to act independently, even though cooperation would lead to a better joint outcome. **It is based on the premise that the firms are not allowed to communicate but are allowed to consider what the other firm is likely to choose, which represents the characteristic of uncertainty in an oligopoly.**

A. A contestable market is one where a monopoly cannot act as a monopoly due to potential threat of competition. This is because entry and exit is easy, as firms can freely enter or leave the market.

B. A kinked demand curve is a phenomenon that occurs in a non-collusive oligopoly which represents the concept of price rigidity arising from mutual interdependence.

C. A principal agent problem occurs when one party (the agent) makes decisions on behalf of another party (the principal), and their interests may not align. This scenario doesn't describe a principal agent problem.

13-C:



- If for example, the firm raises its price, no other firm in the industry will follow, and therefore, this represents “elastic demand” (because, it is the behavior of a firm), and as a result, there is a contraction in quantity demanded by a relatively higher percentage, which results in a “fall in Total Revenue”.

- Importantly, when a firm decreases its price below “P”, all other firms in the industry will follow, and therefore, this represents “inelastic demand” (because, it is the behavior of an industry), and as a result, there will be an extension in qty demanded by a relatively smaller percentage, which results in a “fall in Total Revenue”

→ Therefore, prices remain rigid or unchanged at the “kink”

• Individual firm's expectations about other firms' responses to its changes in prices.

C - The kinked demand curve model of price rigidity in oligopoly is based on the idea that a single firm expects that its competitors will match any price decrease, but will not match a price increase. This leads to a "kink" in the demand curve, as prices are shown to remain unchanged at the kink.

A – Kinked demand curve does not explain collusion; rather it explains the case of a non-collusive oligopoly.

B - A price leader scenario would not result in a kinked demand curve.

D - Barriers to entry would affect the market structure, but not directly explain price rigidity or the kinked demand curve.

14-B:

A – Government provision of market failure represents market failure associated with under-allocation of merit goods.

B – Removal of import quotas is designed to promote free trade and encourage an increase in choices. This has nothing to do with market failure.

C – Price controls on large firms will help to overcome market failure associated with monopoly exploitation.

D – Regulations to limit smoke pollution will help to overcome market failure associated with over-allocation of demerit goods.

15-B:

Marginal Tax Rate:

- **0% up to the tax-free income level**
- **Constant rate (e.g., 20%) above the tax-free income level**

Average Tax Rate:

- **0% up to the tax-free income level**
- **Increasing gradually above the tax-free income level.**

Note 1: Let's use a numerical example to illustrate why the marginal rate of tax and the average rate of tax are 0% up to the tax free income level.

In other words, since no tax is payable on incomes up to the tax-free level, both the marginal tax rate and average tax rate are 0% for this range.

Here's an example to illustrate this:

- Tax-free income level: \$10,000
- Income: \$8,000 (below the tax-free level)

MTR: 0% (no tax on the last dollar earned)

ATR: 0% ($\$0 \text{ tax} \div \$8,000 \text{ income}$)

Note 2:- Let's use a numerical example to illustrate why the average tax rate rises when there is a tax-free allowance (also known as a taxable allowance or basic exemption).

Assume:

- **Tax-free allowance: \$10,000**
- **Tax rate: 20% (constant marginal tax rate)**
- **Income levels: \$15,000, \$25,000.**

Here's how the average tax rate increases as income rises:

1. **Income: \$15,000 (above the tax-free allowance)**
 - **Tax payable: \$1,000 (20% of \$5,000 because the 10,000 earned initially is exempted from tax payment)**
 - **Average tax rate: 6.67% ($\$1,000 \text{ tax} \div \$15,000 \text{ income}$)**
2. **Income: \$25,000 (further above the tax-free allowance)**
 - **Tax payable: \$3,000 (20% of \$15,000 above the allowance)**
 - **Average tax rate: 12% ($\$3,000 \text{ tax} \div \$25,000 \text{ income}$)**

This explains why key B will be correct.

16-C:

Nudge theory seeks to encourage desirable behavior without resorting to force or coercion.

A-. While having a robust tax administration can help deter tax evasion, it's a more traditional enforcement approach and may even be considered to ensure regulation rather than a subtle nudge. Nudge theory aims to influence behavior through gentle, indirect suggestions, rather than direct enforcement or coercion.

B. Imposing heavy penalties is a punitive approach that is classed as a regulation, aimed to discourage tax evasion, rather than encouraging compliance through subtle influences.

C. This option is a classic example of nudge theory in action. By providing information and awareness about the negative consequences of tax evasion, policymakers can gently encourage taxpayers to comply with tax laws. This is why, key C, is correct.

D) It is actually more of a regulatory requirement. Requiring employers to report employee income is a direct mandate, rather than a gentle influence on behavior. Nudge theory aims to encourage behavior through indirect suggestions, rather than direct requirements or coercion.

17-C:

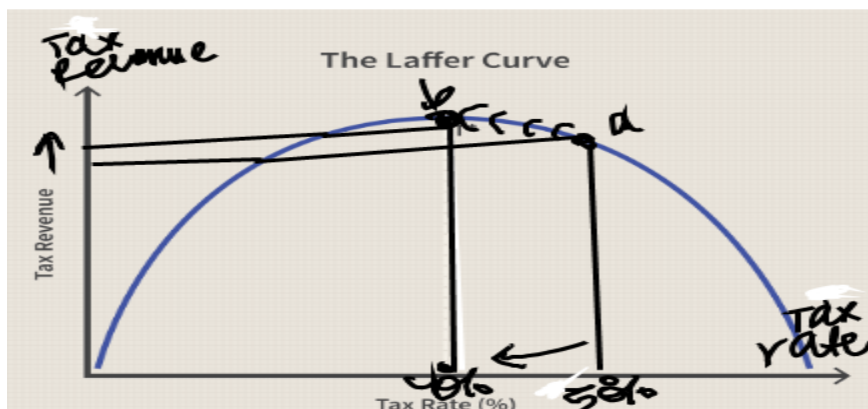
A: It cannot be implied that government policy should aim to keep its level of income tax at its minimum or lowest because that would even bring about a reduction in tax revenues.

B: The observation only relates to income taxes and does not compare them to sales taxes.

C: The observation that reducing high marginal tax rates on richer groups leads to increased tax revenue suggests that the wealthy were previously avoiding or evading taxes due to the high rates. By reducing the rates, the wealthy are more likely to declare their income and pay taxes, resulting in increased revenue.

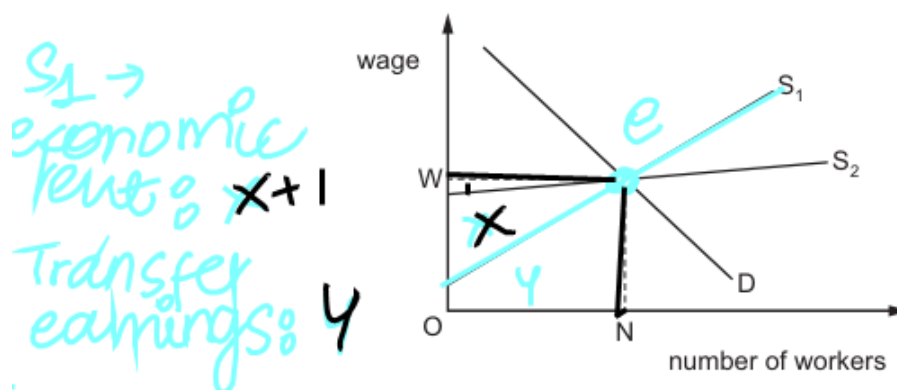
This implies that tax policies aimed at reducing income tax on the rich may lead to increased tax revenue, which can then be used to benefit the poor through social programs or other forms of redistribution.

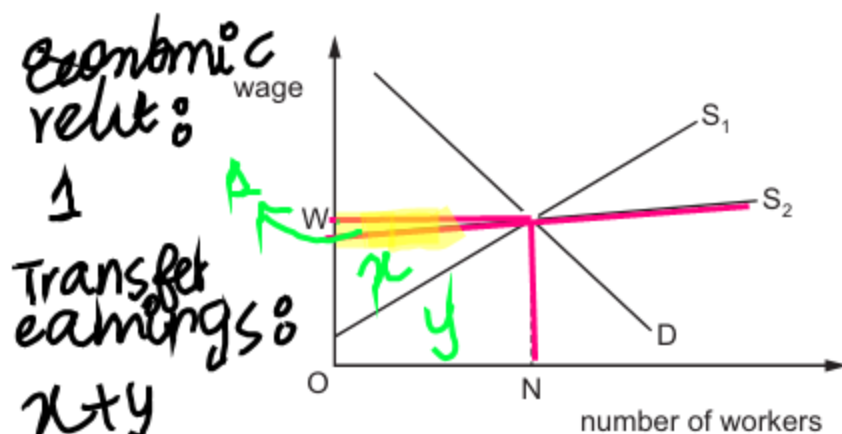
This can be seen in the diagram below.



D: The observation suggests that reducing high marginal rates can lead to increased revenue, but it does not imply that proportional taxes are always superior to progressive taxes.

18-D:





This explains that economic rent falls whereas transfer earnings rise as the supply curve becomes more elastic.

19-C:

A. Adjustments to allow for differences in population size in different countries: Already accounted for in "Real GNP **per head**".

B. Adjustments to allow for differences in rates of inflation in different countries: Already accounted for in "**Real** GNP per head".

C. The hidden economy, also known as the informal sector, can significantly impact Real GNP calculations, as it includes unreported economic activity which might include: dangerous activities such as: drug dealing. By adjusting for the hidden economy, we can get a more accurate picture of the true economic activity and standards of living in different countries. This is why, this becomes correct.

D. Market exchange rates reflect the current exchange rate between currencies, whereas, Purchasing Power Parity (PPP) exchange rates, on the other hand, account for differences in the cost of living between countries, making them a more reliable indicator for comparing standards of living.

20-B:

The combination of decreasing total GDP and rising unemployment, with unchanged discretionary government spending, is most likely to lead to an increase in a country's budget deficit. Here's why:

- ➔ **Decreasing GDP and rising unemployment:** A shrinking economy with a higher degree of unemployment means lower tax revenues for the government from income taxation, indirect taxation and corporation taxation as incomes, expenditures and profits decrease.
- ➔ As unemployment rises, the government faces increased spending on unemployment benefits when unemployment benefit rates remain unchanged as now more people will be unemployed and that too,

automatically as part of automatic fiscal stabilizers, even when discretionary spending remains unchanged.

→ This is why, key B, becomes correct.

21-D:

The effect on national income will be unchanged because:

- The additional police officers are being paid a wage equal to their previous private sector earnings.
- This means that the total amount of money being spent on wages remains the same.
- The only difference is that the government is now paying these wages instead of the private sector.
- As a result, the total national income remains unchanged, as the same amount of money is being spent on wages, just from a different source.

The effect on economic welfare will be uncertain because:

- On one hand, the additional police officers may lead to a reduction in crime, which could increase economic welfare by making people feel safer and more secure.
- On the other hand, the fact that their previous jobs remain unfilled means that there may be a loss of productivity and output in the private sector.
- This could offset the benefits of having additional police officers, leading to an uncertain overall effect on economic welfare

22-A:

In a closed economy, there is no international trade, so there cannot be a balance of trade deficit (or surplus). A closed economy does not engage in trade with other countries.

The other options are all possible in a closed, mixed economy

- B: A budget deficit can occur if the government spends more than it receives in revenue.
- C: Household debt can exist if individuals borrow money to finance consumption or investment.
- D: National debt can exist if the government borrows money to finance its activities.

23-C:

A - "Consumption is a function of the rate of change of income" actually represents the concept of the Marginal Propensity to Consume (MPC), which is the change in consumption in response to a change in income.

B – If a change in investment brings about a change in income, this represents the multiplier effect in action.

C -The Accelerator Principle states that investment is a function of the rate of change of income. This means that when the rate of change in income increases, businesses invest more. This is because changes in income affect the demand for goods and services, and businesses adjust their investment in capital goods accordingly.

D – When investment is affected by interest rates, the concept of marginal efficiency of investment comes in the picture.

24-C:

C: This is an example of seasonal unemployment, as fruit pickers are typically unemployed during the winter months when there is no fruit to pick. Seasonal unemployment is a type of frictional unemployment, which occurs when workers are temporarily unemployed in off peak seasons or when demand for labor falls due to a temporary fall in the demand for the product.

The other options are all examples of structural unemployment, which occurs when workers are unemployed due to a mismatch between their skills and the available jobs, or due to changes in the economy that make their skills obsolete.

A: A car worker replaced by a robot is an example of technological unemployment, a type of structural unemployment.

B: A coal miner whose mine closes due to increased use of solar power is an example of structural unemployment as this is due to a shift in the pattern of demand.

D: A textile worker whose factory closes due to production moving abroad is an example of unemployment due to globalization, a type of structural unemployment.

25-B:

A – A balance of trade surplus represents net injections into the circular flow since export revenues are greater than import expenditures.

B : A government budget surplus represents net leakages in the circular flow since tax revenues are higher than government spending thereby making this correct.

C: A reduction in funds for research and development indicates a fall in injections in to the circular flow and not leakages.

D : Reinvestment of a firm's profits represents investment expenditure which is also an injection into the circular flow.

26-B:

A downward shift of the Aggregate demand curve represents a decrease in Aggregate demand.

A – An increase in firms' stocks of raw materials represents investment by the firms, as a result of which, Aggregate demand will increase.

B – A decrease in export revenues represent a fall in injections into the circular flow and because of this, Aggregate demand will decrease or fall. This is why, key B, becomes correct.

C – A decrease in the marginal propensity to import represents that the marginal propensity to consume increases. This will lead to an increase in consumption and subsequently Aggregate demand will rise.

D – A decrease in savings will bring about an increase in consumption and Aggregate demand.

27-D:

The correct answer is D) devaluation of the currency.

Devaluation of the currency makes exports cheaper and more competitive in the global market, leading to an increase in export demand and subsequently export revenues . Additionally, a devalued currency makes imports more expensive, which can lead to a fall in import expenditures. This results in a BOT surplus in the long-run when the Marshall-Lerner condition is held that is when the sum of elasticities of imports and exports is greater than 1. This will subsequently lead to an increase in Aggregate demand, making this correct.

A) An increase in direct taxation will bring a contractionary effect on Aggregate demand, whereas, a decrease in indirect taxation will bring about an expansionary effect on Aggregate demand. This will leave the overall impact on Aggregate demand uncertain.

B) A relaxation of import controls may lead to an increase in imports that represents a fall in domestic consumption, which could actually decrease aggregate demand.

C) Cuts in public expenditure are likely to decrease Aggregate demand. But cuts in taxes , for example, income tax is likely to increase disposable income that will increase consumption expenditure thereby leading to an increase in Aggregate demand.

28-A:

A – Although households will be spending more, they will also be saving more because they expect themselves to be worse off in the future. As a result of which, growth may be restricted, which means that there may not be a permanent reduction in unemployment. This explains why this will be correct.

B - When households spend their extra disposable income on domestic products, it increases aggregate demand, and because of this production and demand for labor will rise, leading to an increase in employment.

C) When the government's extra spending is on new infrastructure projects, this increases aggregate demand and since because of this new jobs are created, this will lead to a permanent reduction in unemployment.

D) When the government's extra spending is on the salaries of public sector workers, it will increase the purchasing powers of these individuals, which will increase consumption, boost Aggregate demand, and subsequently bring about an increase in employment.

29-D:

The statement describes an increase in the global money supply, which is a key characteristic of an expansionary monetary policy. Central banks are actively increasing the money supply to keep interest rates low, which is a typical strategy to stimulate economic growth and recovery. This is the opposite of a contractionary monetary policy, which would aim to reduce the money supply and increase interest rates.

Expansionary fiscal policy, on the other hand, refers to an increase in government spending or a decrease in taxes to stimulate economic growth, which is not mentioned in the statement.

30-C:

The correct answer is C) the introduction of a basic state pension.

A state pension provides a safety net for the elderly, reducing the need for children to support their parents in old age. This can lead to a decrease in the birth rate, as families no longer feel the need to have many children to ensure financial security in their old age.

Additionally, a state pension can also reduce the household saving ratio, as people are no longer required to save as much for their retirement, knowing that they will receive a basic pension from the state.

The other options are not directly related to reducing birth rates and household saving ratios:

A) Increasing student tuition fees may lead to an increase in the costs of raising children because of which, birth rate may fall. But since this leads to them starting to save from today for their children's education, the savings ratio may rise.

B) Relaxing immigration controls may lead to an influx of new workers and when workers immigrate with their families, this can increase birth rates here. But this may not directly affect savings ratio.

D) Removing tariffs on imported foodstuffs may lead to cheaper food prices, as a result of which, consumption may rise, which may decrease savings ratio. But this has no direct effect on the birth rate.

